

NEO MATHEMATICS CURRICULUM

GreenPrint

Running Curriculum Design Record

Version 1.3 — Reconnection Routes Access and Return Standard

Running design record updated 16 July 2026

“Connection before curriculum. Always.”

Status	Working GreenPrint — live design and production record
Current focus	Age-neutral pathway architecture; repeatable lesson workflow; and mandatory learner access to targeted Reconnection Routes with an explicit return point
Current version	v1.3
Next intended version	v1.4 — completed Foundations unit sequence, pilot evidence and template refinements

Primary sources: NEO Cornerstones in Mathematics; curriculum intent; qualification specifications; learner-access principles; prototype lessons; Director feedback; and GreenPrint versions v0.1–v1.2.

How to Use This Running Design Record

This document is the living record for the NEO KS3–KS4 Mathematics Curriculum build.

Each significant curriculum decision should be added to the design log, with a version number and rationale.

The document should grow in milestone versions rather than being overwritten: v0.1, v0.2, v0.3 and so on.

The GreenPrint is deliberately a working document: a place to test language, capture unresolved questions and preserve decisions.

Version History

Version Date Focus Status

0.1 1 July 2026

Mathematical interpretation of the six

NEO Cornerstones and

initial curriculum architecture Current working version

0.2 July 2026

Stepping Stones

terminology and locked interactive HTML lesson style Superseded by later locked decisions

0.3 July 2026

Learner Guide,

Supporting Adult

Guidance and embedded Scratchpad architecture

Superseded by v0.4

refinement

0.4 4 July 2026

Locked guidance architecture and

refined NEO Maths

Scratchpad interaction standard Current working version

0.5 July 2026

Detailed unit blueprints and production pilot architecture Locked and carried forward

0.6 July 2026

Learner-flow and interactive-entry design rule Locked and carried forward

0.7 July 2026

Nutrition as explicit mathematical nourishment Locked and carried forward

0.8 July 2026 Visual geometry

containment and Locked and carried forward rendering QA

0.9 9 July 2026 Layered Scratchpad

design and QA rules Locked and carried forward

1.0 15 July 2026

Practice Companion

worked-solution unlock rule Locked and carried forward

1.1 15 July 2026

Age-neutral pathways, timetable and lesson ecosystem production standard Current locked version

1.2 16 July 2026 — One-lesson-at-a-time workflow, reference-shell consistency and review gate — Current locked version

Running Design Decisions Log

Decision Rationale Implications Owner / Review Status

Use the six NEO Cornerstones as the primary pedagogical frame for mathematics.

NEO needs one shared whole-school framework rather than parallel frameworks.

Verse-al Ways of

Knowing are retired as lesson headings; their strengths are absorbed into the Cornerstones.

Maths partner + NEO curriculum lead Proposed Build a canonical maths spine with qualification overlays.

A specification describes assessment, not the whole curriculum.

KS3, iGCSE, GCSE,

Functional Skills and

ASDAN can map to the same deeper curriculum outcomes.

NEO + maths partner Proposed

Treat Nutrition

honestly as strongest in applied/contextual mathematics.

It should not be forced into every abstract concept.

Nutrition appears strongly in ratio, proportion, measures, statistics, finance and

Cornerstones Day

projects.

Head of School + maths partner Open question

1. Core Direction

NEO Mathematics should be an interactive, inquiry-led KS3 and KS4 curriculum that keeps the strengths of the Verse-al Maths curriculum, but re-bases the pedagogical structure around the six NEO Cornerstones rather than the five Verse-al Ways of Knowing.

The key shift is:

Verse-al Maths asked: What are the different ways a learner can know mathematics?

NEO Maths asks: How can mathematics help a learner reconnect with learning, self, body, others and the world?

The curriculum should remain grounded in the National Curriculum and qualification pathways, but should not be reduced to exam specifications. The curriculum spine should define what mathematical learning is, while GCSE, iGCSE, Functional Skills and ASDAN requirements sit as overlays.

Curriculum Vision Statement

NEO Mathematics is an interactive, inquiry-led curriculum that helps learners reconnect with mathematics as a language of relationships, change, pattern, fairness, design and decision-making. It is grounded in the National Curriculum and qualification pathways, but not reduced to them. Through the six NEO Cornerstones — Connection, Movement, Creativity, Reflection, Rest and Nutrition — learners explore mathematics as something they can see, move, make, question, pause with and use to sustain real life.

2. The Six NEO Cornerstones Interpreted Mathematically

Connection — Mathematics as relationships Design question: What is connected to what, and how do we know?

Mathematics is the study of relationships: between numbers, quantities, shapes, variables, representations, people and real-world situations. This includes equivalence, ratio and proportion, functions, graphs, sequences, similarity, correlation, networks, mathematical communication, collaborative reasoning and the human story of mathematics.

- Ratio: How do quantities stay connected when they scale?
- Algebra: How can symbols describe relationships?
- Graphs: How can we see a relationship?
- Statistics: What relationships can data reveal, and what might it hide?

Movement — Mathematics as change, transformation and embodied sense-making Design question: What is changing, moving or transforming?

Mathematics describes movement, change, growth, direction, transformation and position. This includes number lines, transformations, coordinates, vectors, gradients, rates of change, sequences, graph behaviour, kinematics, trigonometry and dynamic geometry.

- Number lines: Moving forwards, backwards and across zero.
- Transformations: Shapes moving through reflection, rotation, translation and enlargement.
- Graphs: Stories of movement and change.
- Gradient: How steeply something changes.

Creativity — Mathematics as making, modelling and invention Design question: What can you make with this idea?

Mathematics is not only about finding answers; it is about creating representations, models, conjectures, strategies, proofs and designs. Creativity includes multiple solution methods, pattern creation, mathematical art, modelling, design challenges, coding, data stories and problem posing.

- Algebra: Create rules and formulae for invented situations.
- Sequences: Design growing patterns and describe them.
- Geometry: Construct, tile, tessellate and design.
- Probability: Invent and test games.

Reflection — Mathematics as noticing, reasoning and metacognition Design question: How do I know, and how did my thinking change?

Mathematics requires learners to pause, notice structure, justify decisions, examine errors and become aware of their own thinking. Reflection includes proof, reasoning, error analysis, misconception work, self-explanation, comparing strategies and confidence ratings.

- Algebra: Why is this simplification valid?
- Fractions: What misconception might happen here?
- Graphs: What does this graph make me assume?
- Problem solving: Which strategy helped me get unstuck?

Rest — Mathematics as consolidation, zero, space and sustainable learning Design question: Where is the pause, the space, the balance, or the nothing?

Rest has two roles: as a pedagogical design principle and as a mathematical concept lens. It includes pacing, regulation, chunking, retrieval, low-stakes practice and sensory-load awareness, as well as zero, empty sets, gaps, pauses, balance, limits and intervals.

- Place value: The role of zero.
- Algebra: Balancing equations.
- Geometry: Empty space, area, symmetry and negative space.
- Revision: Retrieval, consolidation and calm fluency.

Nutrition — Mathematics as fuelling life, body, health and sustainable choices Design question: How does this mathematics help us make choices that sustain life?

Nutrition is strongest in applied, contextual, statistical, financial and health-related mathematics. It includes food literacy, health data, budgeting, recipes, scaling, proportional reasoning, sustainable food systems, inequality and access, and decision-making under constraints.

- Ratio: Scaling recipes.
- Percentages: Nutritional labels, sugar content and discounts.
- Statistics: Health data, sleep, food and wellbeing surveys.
- Financial maths: Food budgeting.

3. Proposed NEO Maths Lesson Rhythm

Rather than using the five Verse-al Ways of Knowing, each lesson can use a Cornerstone Cycle. Not every lesson needs all six Cornerstones equally, but every lesson should visibly serve at least one, and most lessons could touch three or four.

Lesson movement Purpose

Connect Start with a relationship, context, image, question, prior idea or learner experience.

Move Use something dynamic: a slider, graph, gesture, number line, transformation, card sort, drag-and-drop or changing example.

Create Learners make something: a rule, model, diagram, conjecture, graph, explanation, design or problem.

Reflect Learners explain, justify, compare strategies, identify a misconception or rate confidence.

Rest Build in pause, consolidation, retrieval, quiet practice or low-stakes checking.

Nourish Apply the maths to a sustaining real-world context where appropriate.

4. Mapping Cornerstones to Mathematical Strands

Maths strand Strong Cornerstones Possible inquiry focus Number Rest, Connection, Reflection What is the structure of number?

What does zero do?

Algebra Connection, Creativity, Reflection How can symbols describe

relationships and hidden rules?

Ratio & Proportion Connection, Nutrition, Creativity How do quantities scale in recipes, maps, budgets and design?

Geometry & Measures Movement, Creativity, Rest How do shapes move, fit, balance and occupy space?

Graphs & Functions Movement, Connection, Reflection How can we see change and relationships?

Statistics Connection, Reflection, Nutrition What stories does data tell, and what stories can it hide?

Probability Creativity, Reflection, Connection What does fairness look like mathematically?

Financial Maths Nutrition, Reflection, Connection How can maths support sustainable choices?

GCSE Problem Solving Reflection, Creativity, Rest How do I choose a strategy and stay regulated when stuck?

5. Recommended Curriculum Architecture

A. Canonical NEO Maths Spine

- Number sense
- Structure and generalisation
- Proportional reasoning
- Spatial reasoning

- Change and functions
- Data and uncertainty
- Mathematical modelling
- Financial and applied reasoning
- Proof, explanation and communication
- Self-regulated mathematical learning

B. Cornerstone Design Layer

- Primary Cornerstone
- Secondary Cornerstone
- Optional contextual Cornerstone
- Fit rating: strong, partial or light
- Sensory load
- Mode: live, asynchronous or independent
- Interactive tool type
- Evidence type
- Mastery descriptors

C. Qualification Overlay

- National Curriculum KS3
- Pearson Edexcel International GCSE Mathematics A
- Foundation/Higher tier
- Functional Skills Maths
- ASDAN or portfolio evidence where appropriate

6. Example Unit Seeds

KS3 Unit Seed: Ratio and Proportion

Driving question: Can maths help us share, scale and sustain things fairly?

- Connection: Ratio connects quantities. Equivalent ratios describe the same relationship.
- Movement: Quantities scale up and down. Learners use sliders, double number lines and bar models.
- Creativity: Learners design recipes, mixtures, maps or scale models.
- Reflection: Learners explain why multiplicative reasoning works and spot additive misconceptions.
- Rest: Calm fluency with equivalent ratios, chunked practice and visual supports.
- Nutrition: Recipes, food labels, value for money, balanced meals and fair access to food.

Possible final task: Design a food pack, recipe or event plan for a chosen number of people. Show ratio reasoning, scaling, cost comparison and a short reflection on fairness or health.

KS4 Unit Seed: Straight-Line Graphs

Driving question: Can a line tell the story of change?

- Connection: Connect equation, table, graph and real context.
- Movement: Gradient describes rate of change; intercept shows starting point.

- Creativity: Learners create a graph story or model a real situation.
- Reflection: Learners explain the meaning of gradient and intercept.
- Rest: Worked examples, graph-reading pause points and low-stakes retrieval.
- Nutrition: Optional applied contexts such as energy use, walking distance, food costs or budgeting.

Possible final task: Create a real-world linear model, explain what the gradient and intercept mean, and judge whether the model is useful or limited.

7. Open Questions for v0.2

1. Should Nutrition be treated as a universal design lens for every mathematical unit, or mainly as an

applied/contextual lens?

2. What is the exact KS3–KS4 unit sequence for Year 9 launch and Year 10 iGCSE Year 1?

3. How should the NEO Mastery scale be phrased specifically for mathematics?

4. Which interactive tool types should be standardised for NEO Maths units?

5. How will portfolio evidence map to qualification overlays and commissioner-facing progress reports?

8. Next Build Step

The next version should develop a full KS3–KS4 curriculum map organised by year group/pathway, mathematical strand, unit title, driving question, primary and secondary Cornerstones, interactive inquiry tool, portfolio evidence, qualification overlay and mastery outcomes.

Running Design Record Update

v0.2 — Lesson Style and Language Lock

Version Update

Version Date Status Summary

v0.2 July 2026 Locked for next

prototype Replaces “Seeds of Change” with “Stepping Stones” and confirms the interactive HTML lesson style for NEO Mathematics.

Locked Design Decision 1 — Use “Stepping Stones” for Learning Outcomes

The term “Seeds of Change” should no longer be used in NEO Mathematics lesson materials. It is too closely associated with the earlier Verse-al framing. The agreed replacement is “Stepping Stones”.

“Stepping Stones” means the core learning objectives or outcomes for a lesson, unit or curriculum node, expressed in a learner-friendly and NEO-aligned way. The phrase suggests gradual progress, re-entry, growth, safety, pacing and movement from one secure point to the next.

Standard usage:

Stepping Stones: use ratio notation; recognise equivalent ratios; solve problems involving direct proportion; use multiplicative reasoning; and explain why quantities must scale by the same factor.

Locked Design Decision 2 — NEO Interactive Lesson Style

NEO Mathematics lessons should use a simple LMS-friendly HTML structure with collapsible panels, a gentle organic colour palette, and embedded interactive activities written directly into the page where possible.

The preferred style is based on the Year 7 interactive Verse-al lesson model, but re-based around the six NEO Cornerstones rather than the five Verse-al Ways of Knowing.

Required HTML Lesson Features

- Begin with a short, inquiry-led Driving Question.
- Use a warm, calm introduction box containing Lesson focus, Stepping Stones and NEO Cornerstones.
- Use collapsible <details> panels for the six NEO Cornerstones: Connection, Movement, Creativity,

Reflection, Rest and Nutrition.

- End with a collapsible Final Task panel that produces portfolio evidence.
- Use a gentle organic colour palette, with a consistent page background and subtly different panel colours.

•Include interactive activities directly in the HTML using learner inputs, selects, buttons, feedback boxes, hints, Socratic prompts and worked solutions.

•Include a separate in-page Practice Companion view, opened by a button from the lesson and returned to by a second button.

- Use trauma-informed design: calm pacing, low-stakes checking, optional support, clear steps and no unnecessary visual clutter.

- Avoid external URLs unless supplied by NEO or the curriculum author.
- Keep the HTML self-contained where possible, using simple inline styles and plain JavaScript.

Required Interactive Pattern

Where an activity is described as interactive, it must do more than display a table or reveal an answer. It should respond to learner input.

- Learner chooses or enters values.
- The page updates calculated outputs or generated content.
- The learner can request a Socratic prompt.
- The learner can request a hint.
- The learner can check an answer or explanation.
- Feedback is displayed in a calm, supportive tone.
- A worked solution is available after an attempt or when pedagogically appropriate.
- Progress is preserved within the current page session where feasible.

Prototype Locked as Example Style

The KS3 Ratio and Proportion lesson prototype is the current reference model for NEO Maths lesson style.

Element Decision Example from prototype Driving Question Short, inquiry-led and age-appropriate.

Can maths help us share, scale and sustain things fairly?

Stepping Stones Use instead of learning objectives/outcomes in learner-facing lesson boxes.

Use ratio notation; recognise equivalent ratios; solve direct proportion problems.

Cornerstone Panels Six collapsible panels aligned to NEO Cornerstones.

Connection, Movement, Creativity, Reflection, Rest, Nutrition.

Interactive Activities Built into the HTML page, not static placeholders.

Recipe Scaler, Ratio Table Builder, Smoothie Design Challenge.

Practice Companion Separate in-page practice mode with progressive questions.

Eight-question practice session with prompts, hints, checks and worked solutions.

Assessment Final portfolio task with mastery rubric.

Emerging, Developing, Secure, Mastering across Understanding,

Application, Communication and

Reflection.

Locked Terminology

Old / Avoid Use Instead Rationale

Seeds of Change Stepping Stones More NEO-aligned; suggests

gradual, safe progress and learner re-entry.

Ways of Knowing NEO Cornerstones NEO Maths is now re-based around the six school-wide Cornerstones.

Learning objectives Stepping Stones Acceptable in technical curriculum mapping, but learner-facing lessons should use Stepping Stones.

Next Build Implication

All future NEO Mathematics lesson prototypes should use the locked style and terminology above. The next curriculum build should apply this format to a small sequence of KS3 and KS4 lessons so the model can be tested for consistency, accessibility and curriculum-vault tagging.

NEO Mathematics Curriculum GreenPrint

Running Design Record v0.3 — Guidance Layers and Embedded

Lesson Scratchpad

Version 0.3 locked design decisions Status: LOCKED for the current NEO Mathematics prototype and to be applied consistently to future lesson builds unless deliberately revised in a later GreenPrint version.

1. Two guidance layers are now standard

Every NEO Mathematics lesson package will include two separate guidance resources because learners and supporting adults need different kinds of orientation.

Resource Primary user Purpose

NEO Mathematics Learner

Guide Learner Short, direct, reassuring orientation; Stepping Stones; a possible route through the lesson; what to try when stuck; useful materials; progress prompts; brief reflection.

NEO Mathematics Lesson

Guidance Teacher, Practitioner, parent, carer or other Supporting Adult Facilitation guidance; mathematical explanation; suggested pacing; prompts; misconceptions; access and regulation; Cornerstone interpretation; evidence and mastery; resources.

The Learner Guide should normally be approximately 2–4 pages. The Supporting Adult Guidance may normally be approximately 5–8 pages. Neither document should assume that the reader is a mathematics specialist.

The standard lesson resource architecture is now:

- Interactive Lesson
- Practice Companion
- Final Portfolio Task
- Learner Guide
- Supporting Adult Guidance

2. Learner Guide language and structure

- What are we exploring? — Driving Question and short orientation.
- Your Stepping Stones — learner-facing progression statements.
- A possible way through — an invitational route, not a rigid lesson sequence.
- When you feel stuck — a short mathematical routine plus the lesson support hierarchy.
- What you might need — useful and optional resources.
- How will I know I am making progress? — accessible 'I can...' indicators without reproducing the

full formal rubric.

- Before you finish — brief reflection on noticing, clarity, revisiting and useful support.

Locked learner-facing principle: 'You can pause. You can return. You do not need to complete

everything in one sitting.'

Locked support principle: 'Using support is part of learning. It is not cheating.'

3. Supporting Adult Guidance structure

- Lesson at a Glance
- Before the Lesson: essential, helpful and optional resources
- Suggested Lesson Journey with invitational timings
- Supporting the Mathematics in plain language
- Common Misconceptions and supportive prompts
- Cornerstone Guidance: mathematical role, what to notice, possible response
- Regulation, Access and Re-engagement guidance
- Evidence and Mastery: Emerging Developing Secure Mastering → → →
- Resources and Further Exploration

Locked pacing principle: timings are invitations, not deadlines. Adults should follow learner

regulation, engagement and mathematical readiness.

4. Guidance access inside every lesson

Every learner-facing HTML lesson should include a Lesson Guides panel near the top of the lesson, immediately after the lesson information block. It should provide two distinct in-page links/buttons:

- Open Learner Guide
- Open Supporting Adult Guidance

The guidance files should open separately so the learner can retain their place in the lesson. Final file paths will be supplied by the NEO platform or content repository and must not be invented during lesson authoring.

Recommended file naming pattern:

- NEO_Maths_[Unit]_[Lesson]_Learner_Guide.pdf
- NEO_Maths_[Unit]_[Lesson]_Lesson_Guidance.pdf

5. Embedded NEO Maths Scratchpad / Interactive Whiteboard

Design direction approved for development: every NEO Mathematics lesson may include an embedded, lesson-local interactive whiteboard provisionally titled the NEO Maths Scratchpad.

The Scratchpad should be native HTML, CSS and JavaScript within the lesson package. It should not send the learner to an external whiteboard URL.

Core purpose: provide an immediate place to sketch, annotate, calculate, draw a diagram, plot rough points, test an idea or record working without leaving the mathematical learning environment.

Minimum Scratchpad feature set

- Open/close or expand/collapse without leaving the lesson.
- Freehand pen input using mouse, touch or stylus where supported.
- Choice of a small number of pen widths.
- Eraser mode.
- Clear board with a confirmation step.
- Undo and redo.
- Optional grid backgrounds relevant to mathematics: blank, square grid and coordinate grid.
- Text note field or simple text tool.
- Save the board state locally during the lesson where the browser environment permits.
- Export the board as an image for possible portfolio evidence.
- Reset or begin a fresh page.
- Accessible labels, large tap targets and keyboard-reachable controls.

Lesson-specific Scratchpad presets

The core Scratchpad should remain consistent across NEO Maths, while each lesson may open with an optional background or template connected to the mathematics.

Lesson Suggested preset Possible uses KS3 Ratio and Proportion Ratio table / double-number-line template Record multipliers; sketch bar models; build equivalent ratios; note recipe calculations.

KS4 Straight-Line Graphs Coordinate grid template Plot points; mark the y-intercept; draw gradient triangles; compare lines; annotate m and c.

Scratchpad placement and interaction model Recommended placement: a persistent 'Open Maths Scratchpad' button near the top of the lesson  and, where useful, smaller 'Open Scratchpad' buttons beside selected interactives. The board should open inside the current page as a large panel or modal-style overlay and close back to exactly the same lesson position.

The Scratchpad is a support space, not a required assessment task. Learners may use paper instead.

A saved screenshot or exported image may be used as evidence where appropriate, but completion of the whiteboard is not itself a mastery criterion.

6. Current reference prototypes

The Ratio and Proportion prototype and the Straight-Line Graphs prototype remain the current examples of the locked NEO interactive lesson style. Future revisions should add the two guidance buttons and the embedded NEO Maths Scratchpad while retaining:

- Driving Question first
- Stepping Stones terminology
- gentle organic palette
- Cornerstone-led collapsible panels
- genuine interactive applications with learner input and immediate support
- Socratic prompt Hint Check Worked solution hierarchy → → →

- in-page progressive Practice Companion
- Final Portfolio Task and Emerging Developing Secure Mastering rubric → → →

7. Next build decision

Build one reusable NEO Maths Scratchpad component, test it inside both current prototype lessons, then create the Learner Guide and Supporting Adult Guidance PDFs for Ratio and Proportion and Straight-Line Graphs. Once tested, the Scratchpad component should become a reusable standard for future lesson HTML.

8. Version 0.4 — Locked Lesson Support and Scratchpad Standard

Status: Locked as the current reference standard for future NEO Mathematics prototype and production lesson builds.

8.1 Standard NEO Mathematics lesson package

Every full NEO Mathematics lesson package should now be designed as a connected set of five resources. The learner must be able to navigate the core lesson independently, while optional support remains immediately available.

Resource Primary user Locked purpose

Interactive Lesson Learner Inquiry-led HTML lesson using collapsible NEO Cornerstone

panels and genuine in-page interactives.

Practice Companion Learner Progressive supported practice

with Socratic prompt Hint → → Check Worked solution. →

Final Portfolio Task Learner Clear product or assignment

linked to Stepping Stones and the Emerging Developing → → Secure Mastering rubric. →

NEO Mathematics Learner

Guide Learner Short, direct and reassuring orientation; Stepping Stones; possible route through; stuck strategies; resources; progress signs; reflection.

NEO Mathematics Lesson

Guidance Teacher, Practitioner, parent, carer or Supporting Adult Mathematical explanation; suggested pacing; prompts; misconceptions; Cornerstone interpretation; access, regulation and evidence guidance.

8.2 Guidance access inside the lesson

Near the top of each HTML lesson, provide a compact support area containing three controls: Open Learner Guide; Open Supporting Adult Guidance; and Open Maths Scratchpad.

Guidance documents should open separately so that the learner's place in the interactive lesson is preserved. The lesson HTML and its PDF guidance files should use stable relative filenames when distributed as a local or platform-hosted package.

Naming pattern: NEO_Maths_[Unit_or_Topic]_Learner_Guide.pdf and NEO_Maths_[Unit_or_Topic]_Lesson_Guidance.pdf.

8.3 NEO Maths Scratchpad — locked design purpose

The NEO Maths Scratchpad is a native, in-lesson HTML canvas and JavaScript thinking space. It must not require an external whiteboard URL or remove the learner from the lesson environment.

Its pedagogical purpose is to support a recurring inquiry cycle: Predict Sketch Test Notice → → → → Revise. The Scratchpad is optional and must never become a compliance or evidence requirement.

Learners may use paper, a physical whiteboard, a notes app or another representation instead.

The Scratchpad may be opened from a persistent lesson-level control and from contextual buttons beside selected interactives. Contextual buttons should load a mathematically relevant working surface where helpful.

8.4 Locked Scratchpad toolbar order

The standard toolbar order is locked as:

Pen Colour Width Eraser Undo Redo Sticky Note Background → → → → → → → This sequence follows the learner's likely workflow: choose how to make a mark, choose its appearance, correct or revise, organise thinking, then choose the mathematical working surface.

The Background label and background dropdown must be grouped as one non-breaking inline control so that they move together when the toolbar wraps. The toolbar should aim to remain on one line on wider screens and wrap cleanly on narrower screens.

8.5 Locked Scratchpad tool specification

Tool / feature Locked behaviour Pen Freehand drawing and handwriting with mouse, touch or stylus input where supported.

Colour A gentle, high-contrast palette: Charcoal, Ocean blue, Moss green, Plum, Earth brown and Berry red. Each stroke retains its selected colour.

Width Thin, Medium and Thick pen widths.

Eraser A dedicated erase mode for removing marks.

Undo / Redo Learners can reverse and restore drawing actions so revising thinking feels normal.

Sticky Note Learners can place multiple editable notes, type a thought/question/reminder, drag notes by the top bar, and delete individual notes.

Background Blank, Square grid, Coordinate grid, Ratio table

and Double number line templates.

New Page Begins a fresh working page after a protective confirmation where existing work may be lost.

Save Working Page Preserves the current working state in browser storage where the hosting environment permits.

Save as Image Exports the Scratchpad canvas as a PNG for possible learner reflection or portfolio evidence.

Clear Board Clears the current page only after confirmation.

Close Closes the modal and returns the learner to the same lesson without external navigation.

8.6 Sticky Notes as a mathematical and metacognitive tool

Sticky Notes are not decorative. They provide a low-pressure way to externalise provisional thinking, questions and reminders without requiring a polished answer.

- “My prediction is...”
- “I think c means...”
- “Check this multiplier.”
- “This model stops working when...”
- “Question for my Practitioner...”

Sticky Notes cycle through a gentle pale palette and remain part of the local Scratchpad working state. They are especially aligned with Reflection, Connection and Rest.

8.7 Lesson-specific Scratchpad presets

The Scratchpad engine remains consistent across NEO Mathematics. Individual lessons may open it with a selected template and a contextual subtitle.

Prototype lesson Preset / contextual surface Inquiry use KS3 Ratio and Proportion Double number line / Ratio table / Blank Predict a scale factor; connect quantities; organise equivalent ratios; sketch a proportional design.

KS4 Straight-Line Graphs Coordinate grid / Blank Predict a graph before reveal; plot clues; draw a gradient triangle; annotate m and c ; sketch a model.

8.8 Current reference prototypes

The current reference implementations are the KS3 Ratio and Proportion interactive lesson v5 and the KS4 Straight-Line Graphs interactive lesson v5. These prototypes define the current visual, interaction and support benchmark for future NEO Maths lesson builds.

- Driving Question first.
- Stepping Stones terminology in learner-facing lesson information.
- Gentle organic lesson palette.
- Collapsible NEO Cornerstone panels.
- Genuine in-page JavaScript interactives with learner input and immediate feedback.
- Contextual Predict Sketch Test opportunities using the embedded Scratchpad. → →
- Progressive Practice Companion with Socratic prompt Hint Check Worked solution. → → →
- Final Portfolio Task with Understanding, Application, Communication and Reflection criteria

across Emerging, Developing, Secure and Mastering.

- Separate Learner Guide and Supporting Adult Guidance PDFs linked from the lesson.
- Embedded NEO Maths Scratchpad using the locked toolbar and tool specification in this section.

8.9 Production and platform testing note

Before the Scratchpad is treated as a production evidence system, its behaviour must be tested inside NEO's final hosting or LMS environment. Browser storage and file export should currently be treated as convenient local working features rather than guaranteed permanent Curriculum Vault storage.

Permanent evidence architecture should be designed separately. The Scratchpad remains first and foremost a learner thinking space.

8.10 Next GreenPrint build priority

Carry the locked v0.4 lesson architecture into the KS3/KS4 curriculum spine and unit map. For each unit, identify Stepping Stones, primary/secondary/optional Cornerstones, fit rating, sensory load, delivery mode, interactive concept, Scratchpad preset, practice evidence and portfolio evidence.

GreenPrint v0.5 - Locked Detailed Unit Blueprint Decisions

Status: LOCKED FOR PROTOTYPE PRODUCTION

This version carries forward GreenPrint v0.4 and locks the session-level blueprint decisions developed for the first two detailed pilot units: Year 9 Pythagoras and the Hidden Length, and Year 10 Straight Lines and Connected Representations.

1. Session-level curriculum design is now a required layer

The NEO Mathematics architecture now distinguishes the unit map from the 45-minute session map.

The required hierarchy is:

- Mathematical Strand -> Unit -> Unit Driving Question -> 45-minute Lesson Sessions -> Stepping Stones.
- One row in the Lesson Architecture Map represents one planned 45-minute learning opportunity.
- A substantial HTML interactive learning environment may support several lesson sessions through different presets, prompts and routes.
- Lesson-session volume and HTML-resource volume must not be treated as the same production measure.

2. Locked Year 9 pilot: Pythagoras and the Hidden Length

The Year 9 Pythagoras blueprint is locked as the reference example of a mathematically distinctive unit arc.

Sequence Session

1 The hidden length 2 Squares on the sides 3 Reconnection: squares, roots and number structure 4 From picture to equation 5 Finding the longest side 6 When the unknown is not the longest side 7 Does the answer make sense?

8 Reconnection: angles and right triangles 9 Is Pythagoras allowed here?

10 Measuring what is hard to reach 11 One diagonal, two routes 12 A diagonal inside a shape 13 From two dimensions towards three 14 What can I now see that was hidden before?

15 How far can indirect measurement take me?

Locked conceptual arc: right triangle -> areas of squares -> relationship -> symbolic equation -> hidden lengths -> theorem selection -> embedded triangles -> indirect measurement.

3. Reconnection routes are attached to the moment of mathematical need

- Reconnection is not a separate catch-up block and is not automatically completed by every learner.
- A Reconnection Route is triggered by a prerequisite signal that is materially blocking current learning.
- Each route records a return point and minimum return evidence.
- The learner repairs enough of the prerequisite connection to resume meaningful progress; they are not required to complete an unrelated intervention programme first.
- The locked Pythagoras pilot routes include squares/roots/number structure and angles/right-triangle recognition.

4. Locked Year 10 pilot: Straight Lines and Connected Representations

The detailed pilot confirms that the coherent mathematical story is relationship -> coordinates -> start -> change -> gradient from points -> equation -> parallelism -> intersection -> conversion -> model limitation -> qualification translation -> synthesis -> model creation.

Design correction locked: set notation and Venn diagrams must not be forced into the Straight Lines inquiry arc. They are to be relocated to a more coherent Data & Uncertainty / Probability representation unit.

5. Unit sequences must follow the mathematics, not a fixed lesson template

- The recurring NEO inquiry rhythm remains a design support, not a compulsory lesson sequence.
- Pythagoras is organised around hidden measurement and area relationships.
- Straight lines are organised around start, change and connected representations.
- Future unit blueprints should identify the mathematical story of the unit before session titles are generated.
- The 68 mapped units must not be auto-expanded using a single repeated 10-lesson or 15-lesson

template.

6. Reusable interactive environment model

The detailed pilots lock the production principle that one interactive environment may support several sessions.

- Pythagoras Explorer is the proposed reusable environment for triangle classification, squares-on-sides, diagram-to-equation matching, hidden-length exploration, missing-side reasoning and theorem selection.

- Dynamic Graph Lab / Straight-Line Graphs Explorer is the reference example of one environment supporting several sessions through different presets.

- Session-specific Driving Questions, initial states, reveals, prompts, Practice Companion focus and Guidance Notes determine the learning experience.

- Production planning must audit reusable tool families before commissioning isolated one-off applications.

7. Mastery checkpoints are not automatically timed end-of-unit tests

The reference mastery session design is a learner-evidence checkpoint. A typical shape is: Arrive 5 / Choose 5 / Evidence 20 / Review 10 / Next Step 5.

- Learners may select evidence from interactive work, Scratchpad images, annotated solutions, verbal or audio teach-back, corrected misconceptions or portfolio work.

- Mastery is judged against the relevant Stepping Stone demonstration.
- The session identifies the next route, including a possible targeted Reconnection Route.
- Qualification assessment remains available where appropriate but is not the sole definition of mathematical learning.

8. Pythagoras production pilot - locked build brief

The next production test is the complete Year 9 Pythagoras unit. The pack should test the following architecture:

- 15 linked 45-minute session lessons.
- A unit index / route page.
- A reusable Pythagoras Explorer visual and interaction language.
- The locked NEO Maths Scratchpad specification from GreenPrint v0.4.
- Contextual Scratchpad presets and Predict -> Sketch -> Test -> Notice -> Revise where mathematically useful.

- Progressive Practice Companion support within each session or linked session environment.
- Learner and Supporting Adult guidance that explains routes, optional Reconnection, pacing, misconceptions and evidence.

- A final Indirect Measurement Portfolio Piece.
- Session files must remain navigable without an external whiteboard or external learning URL.

9. Reference artifacts carried forward

Artifact Current reference

Detailed Unit Blueprint workbook NEO_Mathematics_Detailed_Unit_Blueprints_Y9

_Pythagoras_Y10_Straight_Lines_GreenPrint_v0.

1.xlsx Year 9 / Year 10 Lesson Architecture Map NEO_Mathematics_Year9_Year10_Lesson_Architecture_Map_GreenPrint_v0.1.xlsx

Curriculum Vault Map NEO_Mathematics_KS3_KS4_Curriculum_Vault_

Map_GreenPrint_v0.1.xlsx Reference prototype lesson NEO_Maths_KS3_Ratio_Proportion_Interactive_v5.html
Reference prototype lesson NEO_Maths_KS4_Straight_Line_Graphs_Interactive_v5.html

10. Learner-flow and interactive entry design rule

The Straight-Line Graphs repo migration pilot established that a mathematically strong interactive is not sufficient on its own. A learner may still be left in the fog if the lesson does not make clear why the activity appears at that point, what action is expected, what mathematical feature should be noticed, and how the resulting idea connects to the next stage of learning.

- Locked design rule: Every interactive must have a clear entry instruction, a stated mathematical noticing goal, and an explicit conceptual bridge to the next stage of the lesson.
- Locked sequence rule: Learners should encounter and experience an idea before they are asked to recognise, apply, compare, justify or create with it. Activities must be ordered by conceptual dependency, not simply by Cornerstone category or by the order in which they were originally built.
- Interactive entry instruction: Tell the learner concretely what to change, choose, drag, enter, compare or build. Activity titles must not be expected to carry the instructions.
- Mathematical noticing goal: State what relationship, invariant, change, structure or evidence the learner is trying to notice. This should be visible before or during the investigation, not only in feedback after completion.
- Conceptual bridge: Close each substantial investigation by naming or eliciting the idea that the learner should carry into the next activity. The next panel should visibly build from that idea.
- Cornerstone clustering remains the default layout rule where repeated Cornerstones occur, but conceptual dependency takes precedence. Repeated panels may be interleaved only when returning to a Cornerstone is mathematically necessary for the learning journey.
- Reference example: Plotting Straight Line Graphs migration pilot. The stronger sequence

became: experience one relationship through equation-table-graph connection -> recognise the same relationship when representations are separated -> deliberately change m or c and compare before/after -> reconstruct a line from evidence -> create a meaningful model.

- New migration QA questions: Why is the learner doing this now? What exactly are they expected to do? What mathematics are they trying to notice? What idea should they carry forward next?

11. Version history update

v0.6 locks the Learner-Flow and Interactive Entry Design Rule established through the Straight-Line Graphs repo migration pilot: every interactive requires a clear entry instruction, a stated mathematical noticing goal and an explicit conceptual bridge to the next stage. It also locks conceptual dependency as the governing sequence principle, with Cornerstone clustering remaining the default unless the mathematics requires an intentional return.

v0.5 locks the two detailed pilot blueprints, the separation of set notation from the Straight Lines unit, the session-level hierarchy, targeted Reconnection Routes, mathematically distinctive unit arcs, reusable interactive environments and the complete Pythagoras production pilot brief.

GreenPrint v0.7 - Nutrition as Explicit Mathematical Nourishment

Status: LOCKED FOR PROTOTYPE PRODUCTION

This version carries forward GreenPrint v0.6 and resolves the Mathematics interpretation of the NEO Nutrition Cornerstone. Nutrition remains one of the six explicit NEO Cornerstones and must be referenced in every NEO Mathematics lesson. The lesson must identify honestly what form of nourishment is authentic to the mathematics.

1. Locked Nutrition principle

- Locked design rule: Every NEO Mathematics lesson makes explicit reference to the Nutrition Cornerstone. Nutrition may be expressed through authentic contextual nourishment or through intellectual nourishment.

- Nutrition design question: What is this learning nourishing - practically, physically or intellectually?

- Non-contrivance rule: A food, body, health or sustainability context must not be manufactured merely to populate a Nutrition panel.

- Intellectual nourishment rule: Where contextual nourishment is not authentic, the lesson should identify the mathematical capacity, habit, curiosity, confidence, agency or sense of coherence that the learning is strengthening.

2. Two legitimate Nutrition modes

- Mode A - Authentic contextual nourishment

Use where the mathematics genuinely lives in food, hydration, the body, health, physical self-care, sustainable resources, food security or other forms of practical nourishment. Examples include recipe ratio, health statistics, food budgeting, resource allocation and sustainability.

- Mode B - Intellectual nourishment

Use where the mathematical content does not require a literal nourishment context. The learner is invited to notice what the learning is feeding or strengthening in their mathematical thinking, such as clarity, pattern awareness, precision, curiosity, prediction, evidence checking, confidence, agency, tolerance of uncertainty or appreciation of structure.

3. Required learner-facing function of a Nutrition panel

- Name honestly which Nutrition mode is being used.
- Identify the mathematical habit, capacity or practical form of nourishment being strengthened.
- Give the learner a low-pressure way to notice what feels stronger or clearer.
- Ask the learner to identify one idea, habit or connection worth carrying forward.
- Avoid turning Nutrition into a generic wellbeing reflection or a disguised assessment score.

4. Nutrition as a whole-curriculum design condition

In addition to its explicit lesson reference, nourishment is a non-negotiable quality condition for the whole NEO Mathematics curriculum. A lesson should leave the learner with something intellectually sustaining: greater clarity, stronger sense-making, a meaningful success, a new connection, increased agency, deeper curiosity or a more coherent mathematical model.

The GreenPrint QA question is: How is this lesson nourishing the learner mathematically? If the lesson is active but does not reduce fog, reveal structure, create meaningful agency or strengthen a worthwhile mathematical habit, the pedagogy requires review.

5. Reference examples

- Place Value migration - intellectual nourishment. The lesson explicitly nourishes attention to detail, pattern awareness, confidence with crowded numbers and sense-checking. The learner identifies one place-value idea worth carrying forward.
- Plotting Straight Line Graphs repo migration - intellectual nourishment. The lesson explicitly nourishes noticing change, connecting representations, prediction, evidence checking and algebra confidence. The learner identifies one idea worth carrying into the next lesson.
- Ratio and recipe scaling - authentic contextual nourishment may be appropriate where the mathematical inquiry genuinely concerns food, sharing, sustainability or feeding a group.

6. Translation Matrix and repo-migration implication

- Nutrition can no longer be recorded as absent solely because a literal food or health context is

weak.

- Each migrated lesson must record Nutrition mode: contextual nourishment or intellectual nourishment.
- The Translation Matrix should capture what is being nourished and the learner-facing carry-forward prompt.
- Existing repo lessons that force Nutrition through a decorative context should be reviewed and translated into the authentic mode that best fits the mathematics.

7. Version history update

v0.7 locks Nutrition as an explicit Cornerstone reference in every NEO Mathematics lesson and defines two legitimate modes: authentic contextual nourishment and intellectual nourishment. It also establishes nourishment as a whole-curriculum quality condition. Place Value and Plotting Straight Line Graphs are the first reference examples of intellectual nourishment.

v0.6 locks the Learner-Flow and Interactive Entry Design Rule established through the Straight-Line Graphs repo migration pilot.

GreenPrint v0.8 - Visual Geometry Containment and Rendering QA

Status: LOCKED FOR PROTOTYPE PRODUCTION

This version carries forward GreenPrint v0.7 and formalises a visual-geometry production gate established during the Pythagoras repo-migration pilot. Mathematical correctness is not sufficient when part of the intended construction is clipped, hidden or visually cut off. Diagrams and dynamic geometry environments must be checked as complete visual objects before a lesson is handed over.

1. Locked Visual Geometry Containment rule

- Locked design rule: Every diagram, graph, coordinate environment or dynamic geometry application must be checked against the full intended construction, not only the central mathematical object.
- Full-construction rule: The visual bounds must include attached shapes, constructed squares, labels, right-angle markers, annotations, draggable handles, arrows and any other element that forms part of the learner's evidence.
- Safety-margin rule: Visual bounds must include deliberate breathing room on every side. A construction should not merely fit mathematically; it should remain comfortably readable.
- Default-state check: The complete construction must be visible when the learner first opens the activity or panel.
- Interaction-extremes check: Where the learner can drag, resize, rotate, scale, randomise or otherwise transform a construction, the intended interaction range must be tested for clipping and loss of labels or controls.
- Container-and-viewBox rule: For SVG and canvas-based geometry, the application container and

drawing coordinate system must be dimensioned from the bounding box of the whole intended construction. Cropping to the triangle, graph or central shape alone is not acceptable where attached geometry extends beyond it.

2. Required pre-delivery containment QA

Step 1: Identify the complete intended construction and list all elements that must remain visible.

Step 2: Determine or calculate the minimum and maximum visual bounds of those elements.

Step 3: Add a deliberate visual safety margin to the calculated bounds.

Step 4: Check the default learner view for clipping, crowding and labels touching container edges.

Step 5: Exercise the full intended interaction range, including extreme draggable positions, random states and larger generated objects.

Step 6: Check that learner instructions and the evidence source remain visible together where the task requires comparison.

Step 7: Repeat the visual check after any later geometry, labelling, scaling or layout edit.

3. Visual-geometry shipping gate

A visual mathematics lesson is not QA-complete until the reviewer can answer YES to all of the following:

- Can I see the whole mathematical construction without scrolling inside the drawing area?
- Are all attached shapes and constructed objects fully visible?
- Are labels, markers and handles inside the visual bounds and comfortably separated from the

edges?

- Does the construction remain contained at the intended interaction extremes?
- Can the learner see the diagram or graph that the instructions ask them to use as evidence?
- Has the visual been rechecked after the most recent geometry or layout change?

4. Reference example: Pythagoras Reflection diagram

In the Pythagoras' Theorem repo-migration pilot, the Reflection section was pedagogically improved by adding a labelled right-angled triangle and the three squares as an explicit evidence source. The first render nevertheless clipped the upper part of the square on the hypotenuse because the SVG viewBox had been sized around the central triangle rather than the complete construction.

The correction expanded the drawing bounds to include all three attached squares plus a deliberate safety margin. The underlying lesson learned is now locked: when geometry is constructed on a mathematical object, the QA bounding box belongs to the construction as a whole.

5. Relationship to the Learner-Flow rule

Visual containment is part of learner flow. An instruction such as 'use the diagram below as your evidence' fails pedagogically if the evidence is partly cropped. Clear entry instructions, mathematical noticing goals and conceptual bridges therefore require an equally clear and complete visual evidence source.

6. Version history update

v0.8 locks the Visual Geometry Containment and Rendering QA rule: complete-construction bounds, deliberate safety margins, default-state checks, interaction-extremes checks and a pre-delivery visual shipping gate are now required for diagrams, graphs and dynamic geometry applications.

v0.7 locks Nutrition as an explicit Cornerstone reference in every NEO Mathematics lesson and defines authentic contextual nourishment and intellectual nourishment.

v0.6 locks the Learner-Flow and Interactive Entry Design Rule.

GreenPrint Update v0.9 — Scratchpad Design Rules

Date: 2026-07-09

This update locks in the newly agreed Scratchpad design features following the Pythagoras lesson migration and Scratchpad refinement work.

Purpose Lock in the newly agreed Scratchpad design features so that future NEO Maths lessons provide a more reliable, mathematically helpful and learner-friendly working space.

New Design Rules to Add

1. Layered Scratchpad Architecture — The Scratchpad should use separate layers for: (a) a persistent

mathematical background layer, (b) a learner-ink layer for pen, line and eraser actions, and (c) an annotation layer for sticky notes or movable thinking objects. Rationale: this prevents mathematical supports from being damaged when the learner erases their own marks.

2. Persistent Background Rule — When the learner erases a line, sketch or mark, the selected

background must remain intact. The Eraser should affect learner marks only, not the background.

3. Straight-Line Tool Rule — The Scratchpad should include a dedicated straight-line tool as well as

freehand pen input. Expected interaction: choose the Line tool; click or press to set a start point; drag to preview the line; release to place the line. The line should respect the currently selected colour and width.

4. Working History Rule — All learner-created objects and marks, including straight lines, should

participate fully in Undo, Redo, saved working-page state, clear-board behaviour and image export.

5. Clear Board Rule — Clear board should remove learner ink, line objects and sticky notes, but

preserve the selected mathematical background.

6. Background Change Rule — If the learner changes the background, their existing learner marks

should remain unless they explicitly choose to clear them.

7. Export Rule — When the Scratchpad is saved as an image, the export should combine the visible

background layer with the learner's marks and notes so that the mathematical context is preserved.

8. Toolbar Clarity Rule — The Scratchpad toolbar should present tools in a stable, comprehensible

order. Current agreed order: Pen Line Colour Width Eraser Undo Redo Sticky Note → → → → → → → → Background.

9. Learner Instruction Rule — Where a tool is less obvious, a brief status message or prompt should

explain what the learner is expected to do. Example: for the Line tool, explain that the learner should press at the start point, drag to the end point, and release.

10. Scratchpad QA Rule — Before any lesson using the Scratchpad is delivered, check that: the

background persists after erasing; the Line tool works cleanly and draws truly straight lines; Undo and Redo work for all tool types; Clear board preserves the background; exported images include the background and learner marks; and the Scratchpad behaves consistently across the lesson.

Implementation Note

These rules should now be treated as part of the running NEO Mathematics GreenPrint and used as defaults in future lesson builds and lesson migrations.

GreenPrint Update v1.0 — Practice Companion Worked-Solution Unlock Rule

Date: 2026-07-15

Purpose Lock the agreed Practice Companion safeguard so worked solutions support learning rather than replacing learner reasoning. The Practice Companion remains a guided practice environment, not an answer-reveal tool.

Locked Behaviour

In every NEO Mathematics Practice Companion, the Worked solution button must begin greyed out and disabled. It should unlock only after the learner has used the Socratic prompt, used the Hint, and had two tries at the question.

Rationale This preserves productive struggle while still providing safety and support. Learners are not abandoned: they can access a prompt, then a hint, then attempt the problem. The worked solution becomes a recovery and learning tool after effort, not a shortcut before thinking.

Required UI Pattern

The Practice Companion should show: Return to lesson; progress dots; question card; answer box; Socratic prompt; Hint; Check; disabled Worked solution; Scratchpad; Previous and Next navigation.

A short learner-facing note should explain when the worked solution unlocks.

Unlock Logic

For each practice question, track three conditions separately: prompt used, hint used, and number of checked attempts. Unlock the Worked solution only when all three are true: prompt used = true, hint used = true, attempts ≥ 2 . The rule resets per question, not globally across the whole companion.

Feedback Rule

If the learner clicks the Worked solution button while it is locked, the interface should explain exactly what remains to do, for example: “Worked solution unlocks after you use the Socratic prompt, use the hint, and have two tries.”

Accessibility and Regulation Note

This rule should be implemented gently. The button should look unavailable but the learner should be told why. The prompt/hint/tries sequence should reduce anxiety by making the support path visible and predictable.

QA Rule

Before delivery, test at least one Practice Companion question by confirming: the Worked solution button is initially disabled; it remains disabled after prompt only; it remains disabled after hint only; it remains disabled after one try; it unlocks after prompt + hint + two tries; navigating to another question restores that question's own lock state.

Version History Update

v1.0 locks the Practice Companion Worked-Solution Unlock Rule. v0.9 locked the Scratchpad Design Rules. v0.8 locked Visual Geometry Containment and Rendering QA. v0.7 locked Nutrition as an explicit Cornerstone reference in every lesson. v0.6 locked Learner-Flow and Interactive Entry Design.

GreenPrint Update v1.1 — Age-Neutral Pathways and Lesson

Ecosystem Production Standard

Date: 2026-07-15

Status: LOCKED

This update carries forward every earlier locked GreenPrint decision and establishes the curriculum architecture, planning envelope and production standard to be used for future NEO Mathematics development unless deliberately revised in a later version.

1. Purpose and Direction of Travel

NEO Mathematics will be an age-respectful, interactive and responsive curriculum that meets learners at their current mathematical point without defining them publicly by chronological school year. The curriculum remains mapped internally to National Curriculum and qualification expectations, but learner-facing resources should foreground mathematical journey, connection and progress.

The immediate direction is to continue building the current Pythagoras sequence lesson by lesson, using evidence from each lesson to decide how much teaching and practice the next mathematical connection requires. The same production model will then support the Intermediate pathway and, after separate qualification-hour planning, the Advanced pathway.

- Connection before curriculum remains the governing principle.
- Age, school year and pace must not become public judgements about a learner's mathematical

identity.

- The curriculum spine remains canonical; qualifications remain overlays rather than the whole curriculum.

- Stable lesson architecture will reduce cognitive load, while content volume and route remain responsive.

- Topic coverage will be secured through a live coverage map rather than a rigid lesson-count allocation.

2. Age-Neutral Learner Pathways

The following pathway language is now the standard learner-facing architecture. Internal year-group and qualification mappings remain available for curriculum planning, reporting, audit and file management.

Learner-facing pathway Internal mapping Primary purpose Display rule Foundations

Broadly Years 7–9, with

bridges and reconnections as required Secure the conceptual structures needed before qualification study

Use NEO Mathematics in

lesson titles; show pathway in navigation or course maps where useful

Intermediate Broadly Years 10–11 IGCSE Mathematics preparation

and qualification Do not confuse the NEO pathway with Foundation/Higher or Core/Extended examination tiers

Advanced Broadly Years 12–13 AS and A-Level Mathematics

Qualification route and guided-hour model must be stated separately

Locked learner-facing label rule

Do not display Year 7, Year 8, Year 9, Year 10, Year 11, Year 12 or Year 13 in learner-facing lesson titles, headers or guidance documents. Internal filenames and metadata may retain year codes where they are useful for curriculum management.

The plural label Foundations is preferred to Foundation Level because Foundation Tier already has a specific meaning in Pearson IGCSE. Examination tier must always be tracked separately from NEO pathway placement.

3. Timetable and Guided-Learning Planning Envelope

The normal Foundations and Intermediate planning model is three 45-minute guided lessons each week over a 38-week academic year.

Pathway Lessons/

week Weeks/ year Lessons/ year Hours/ year Locked planning position

Foundations 3 × 45 min 38 114 85.5

Normal annual model; bridges and reconnections use the same lesson envelope

Intermediate 3 × 45 min 38 114 85.5 Two-year total: 228 lessons / 171

guided hours

Advanced To be

confirmed To be confirmed — — Must be planned separately against the selected AS/A-Level qualification

Planning evidence: Pearson describes International GCSEs as two-year programmes with 120 guided

learning hours; Cambridge IGCSE Mathematics is designed around approximately 130 guided learning hours. The Intermediate allocation of 171 guided hours therefore provides a deliberate buffer for reconnection, slower processing, consolidation, regulation, assessment and interrupted attendance.

Advanced caveat: Cambridge planning guidance is approximately 180 guided hours for AS Level and

360 guided hours for the complete A Level. The Foundations/Intermediate timetable must not therefore be copied automatically into Advanced provision.

Timings are invitations, not deadlines

The 45-minute shape provides orientation and rhythm. A learner may pause, return or continue across another session. Regulation and mathematical readiness take precedence over completing every screen or activity within one sitting.

4. Responsive Topic and Lesson Allocation

A fixed number of lessons will not be assigned permanently to every topic in advance. The 114 annual lessons form a planning envelope, not a demand to distribute every available session before teaching begins.

- Define the topic's essential mathematical connections, Stepping Stones and qualification coverage.
- Use prerequisite evidence or a short diagnostic to identify likely Reconnection Routes.
- Build an initial short cluster of lessons rather than the whole topic at once.
- Review lesson evidence, Practice Companion performance and learner regulation before fixing the

next cluster.

- Extend, consolidate or conclude the topic according to conceptual security, not calendar pressure

alone.

- Maintain a live coverage map so responsive pacing never produces accidental specification gaps.
- Preserve deliberate annual capacity for reconnection, consolidation, assessment, portfolio

evidence and disruption.

A topic is complete when its essential Stepping Stones have been encountered, practised, connected and evidenced to the required level, and when the curriculum team can identify any remaining Reconnection Routes explicitly.

5. The Standard NEO Mathematics Lesson Ecosystem

Every completed NEO Mathematics lesson package should be understood as a connected ecosystem rather than a single HTML page.

Resource Primary role Non-negotiable relationship Interactive Lesson Inquiry, connection, exploration and evidence Acts as the hub and links directly to all learner supports Resource Primary role Non-negotiable relationship Practice Companion Graduated supported practice Uses the lesson's mathematical language and preserves the worked-solution safeguard

NEO Maths

Scratchpad Optional thinking and representation space Opens without removing the learner from the lesson or Practice Companion Learner Guidance Calm orientation and self-navigation Explains the lesson route and support pathway without reproducing answers

Supporting Adult

Guidance Facilitation, access and mathematical support Protects learner agency and explains how to support without taking over

Portfolio / Evidence

Task Meaningful evidence where appropriate May draw on lesson work, Scratchpad exports, explanations or corrected thinking

6. Interactive Lesson Structure and Format

6.1 Learner-facing opening

- NEO Mathematics title with no chronological year label.
- Short inquiry-led Driving Question.
- Warm orientation block containing lesson focus, Stepping Stones and NEO Cornerstones.
- Compact support area near the top: Open Learner Guidance; Open Supporting Adult Guidance;

Open Maths Scratchpad.

- Clear entry instruction, mathematical noticing goal and conceptual bridge for every interactive.

6.2 Default 45-minute Cornerstone rhythm

Approximate phase Invitation Purpose

Connect — 15 minutes Relationship, context, image, question or prior idea Establish meaning and activate a secure entry point Move — 15 minutes Dynamic example, gesture, drag, transformation, graph or model Make change and structure perceptible Reflect / Create — 7 minutes Explain, compare, construct, conjecture or model Make thinking visible and generative Rest / Nourish — 3 minutes Pause, regulate, retrieve or name mathematical nourishment Support sustainable learning

Evidence — 5 minutes Self-review, explanation, check or saved evidence Identify current security and next connection Not every lesson must weight the six Cornerstones equally. Every lesson must reference all six honestly and identify whether Nutrition appears through authentic context, intellectual nourishment or a brief sustaining-learning connection.

6.3 Interactive and feedback standard

- An interactive must respond meaningfully to learner input; a static reveal is not sufficient.
- Use progressive disclosure, collapsible panels and calm visual hierarchy to control cognitive load.
- Feedback must be dynamically determined from the learner's input wherever mathematically

possible.

•Equivalent correct answers, valid numerical tolerances and alternative representations must be accepted rather than relying on one hard-coded answer string.

- Incorrect feedback should identify a useful next action without revealing the complete method.
- Progress should remain available within the current session where feasible.
- The learner may pause, return and use paper or another representation at any point.

7. Practice Companion Structure and Format

7.1 Question count

Each Practice Companion will contain between 4 and 10 questions. The number is decided lesson by lesson according to cognitive weight, variety of representation and the amount of practice genuinely needed.

Typical range Use when 4–5 questions Problems are multi-step, diagram-rich, unfamiliar, construction-based or explanation-heavy 6–7 questions A normal balance of recognition, application, reasoning and transfer is needed 8–10 questions Items are shorter or designed for fluency, retrieval and carefully varied examples Where a companion contains a larger set, questions may be labelled Core Practice and Explore Further so the learner does not experience every item as a compulsory hurdle.

7.2 Required question architecture

- Question card with clearly separated information, diagram and response area.
- Measurements and labels positioned clearly away from diagram lines and visual edges.
- No unnecessary information that gives away the method; shape properties may need to be

inferred.

- At least one opportunity for transfer, explanation, diagram construction or interpretation where

appropriate.

- Socratic prompt, Hint, Check, Worked solution and Scratchpad access for every question.
- Previous/Next navigation and visible question progress.
- Calm, dynamic feedback that accepts mathematically equivalent answers.

7.3 Universal worked-solution release rule

Locked rule for every question

The Worked solution button begins visible but greyed out and disabled. It unlocks only when all three conditions are true for that question: Socratic prompt used; Hint used; and at least two answers entered and checked. The conditions reset independently for each question.

Recommended learner pathway: Think and sketch Socratic prompt first try Hint second try → → → →

→ Worked solution.

If a learner selects the locked button, the interface must state exactly what remains to do. A Supporting Adult must not provide the complete worked solution verbally while the button is locked.

7.4 Practice Companion QA

- Test the initial locked state and each partial condition separately.
- Confirm unlocking after prompt + hint + two checked attempts.
- Confirm that each question retains its own lock state during navigation.

- Test every accepted correct-answer form and relevant tolerance.
- Test diagrams at default and responsive sizes, including interaction extremes.
- Check that feedback never misses a correct mathematical equivalent or reveals an answer too

early.

8. NEO Maths Scratchpad Structure and Format

The Scratchpad is an optional, lesson-local thinking space. It supports the recurring inquiry cycle Predict Sketch Test Notice Revise and must never become a compliance or mastery → → → → requirement.

- Native HTML, CSS and JavaScript within the lesson package; no external whiteboard dependency.
- Opens and closes without losing the learner's position in the lesson or Practice Companion.
- Separate persistent background, learner-ink and annotation layers.
- Toolbar order: Pen Line Colour Width Eraser Undo Redo Sticky Note → → → → → → → →

Background.

- Backgrounds may include blank, square grid, coordinate grid, ratio table, double number line or

lesson-specific surfaces.

- Eraser and Clear Board remove learner-created marks while preserving the selected mathematical

background.

- Undo, Redo, local save and image export include all learner-created objects and straight lines.
- Export combines background, marks and notes so the mathematical context remains visible.
- Large tap targets, keyboard-reachable controls, accessible labels and brief tool instructions are

required.

- Paper, a physical whiteboard, a notes app or another representation always remains a valid

alternative.

Browser storage and image export are local conveniences until the final NEO hosting and evidence architecture has been tested. They must not be described as guaranteed permanent Curriculum Vault storage.

9. Learner Guidance Structure and Format

Each lesson package includes a separate learner-facing guide, normally approximately 2–4 pages, written directly to the learner in calm, respectful language. The guide must not assume a particular age or that the learner can complete the lesson in one sitting.

- NEO Mathematics title with no school-year label.
- Welcome / what we are exploring.
- Stepping Stones or what you are learning.
- Before you begin: useful and optional resources and regulation choices.
- A possible route through the lesson, aligned to the Cornerstone rhythm.
- Lesson-specific diagram, representation or interaction guidance.
- Practice Companion support pathway and exact worked-solution release conditions.
- When you feel stuck: one small mathematical or regulation action at a time.

- How you may show what you understand and a short self-review.
- Optional space for notes or sketches where the format benefits from it.

Locked learner-facing principles

You can pause. You can return. You do not need to complete everything in one sitting.

Using support is part of learning; it is not cheating. The guide must never reproduce the lesson's worked answers.

10. Supporting Adult Guidance Structure and Format

Each lesson package includes a separate Supporting Adult Guidance document, normally approximately 5–8 pages. It must be usable by a teacher, practitioner, parent, carer or other adult who is not a mathematics specialist.

- Lesson purpose and mathematical focus at a glance.
- What the learner is developing.
- The adult's role: protect thinking, agency and regulation rather than take over.
- Before the lesson: essential, helpful and optional preparation.
- Support through the Cornerstone lesson phases with invitational timings.
- Short questions that preserve learner thinking and allow processing time.
- Plain-language explanation of important mathematical connections.
- Lesson-specific misconceptions, difficulties and supportive responses.
- Practice Companion pathway and the prohibition on giving the worked solution while locked.
- Regulation, access, re-engagement and signs of overload.
- Evidence, self-review and a valid finishing point.
- Optional observation and reconnection notes.

The adult guide must use age-neutral language, avoid deficit framing and treat incomplete work as information for the next connection rather than automatic failure.

11. Visual, Accessibility and Containment Standards

- Warm organic palette, readable typography, strong contrast and restrained visual density.
- Instructions chunked into one clear action at a time where possible.
- Choice of spoken, typed, drawn, calculated or annotated evidence.
- No time-pressure language; timings are invitations.
- Diagrams sized from the complete construction, including attached squares, labels, handles and

movement range.

- Deliberate safety margin around every visual construction.
- Default state and interaction extremes tested for clipping, crowding and label loss.
- Measurements placed clearly away from sides, boundaries and other labels.
- Responsive layout checked at intended screen sizes and touch targets kept comfortably usable.
- Avoid unnecessary external links and keep lesson packages self-contained where possible.

12. Production, Naming and Package Standard

Learner-facing titles remain age-neutral. Internal filenames may retain curriculum-stage codes so the production team can map resources accurately.

Artifact Internal naming pattern

Interactive Lesson `NEO_Maths_[InternalStage]_[Unit]_[Lesson]_vX.Y.html`

Learner Guidance `NEO_Maths_[InternalStage]_[Unit]_[Lesson]_Learner_Guide_vX.Y.pdf`

Supporting Adult

Guidance `NEO_Maths_[InternalStage]_[Unit]_[Lesson]_Supporting_Adult_Guidance_vX.Y.pdf`

Lesson Package `NEO_Maths_[InternalStage]_[Unit]_[Lesson]_vX.Y_Package.zip`

- The package index must identify every included artifact.
- Lesson buttons must open the guidance PDFs and Scratchpad from inside the same package or approved repository structure.
- Internal stage codes such as Y9 may remain in filenames, metadata and curriculum maps but not in learner-facing page furniture.
- Every change receives a version increment and a concise version reason.
- Previous versions remain recoverable through version history rather than being silently overwritten.

13. Production QA Shipping Gate

A lesson ecosystem is ready for delivery only when the reviewer can answer YES to all of the following:

- Are learner-facing titles and headers free from chronological year labels?
- Are the Driving Question, Stepping Stones and mathematical noticing goal clear?
- Does every interactive respond meaningfully and give mathematically valid dynamic feedback?
- Are complete diagrams visible, readable and contained at default and interaction extremes?
- Does the Practice Companion contain an appropriate 4–10 questions rather than a fixed quota?
- Does every practice question enforce prompt + hint + two attempts before the worked solution unlocks?
- Does the Scratchpad preserve backgrounds, support all agreed tools and remain optional?
- Do both guidance documents match the lesson, use age-neutral language and preserve the support pathway?
- Can the learner pause, return and use an alternative way of recording thinking?
- Have the HTML, JavaScript, responsive layout, PDF links and package navigation been tested after the most recent change?

14. Retrospective Documentation Update Required

Earlier curriculum and production documents must now be audited so learner-facing year labels and fixed allocation assumptions do not conflict with v1.1. The mathematical mapping itself should be preserved internally.

Document / artifact Required change Preserve internally

Detailed Unit Blueprint

workbook Add Foundations / Intermediate / Advanced pathway field; replace fixed lesson totals with provisional ranges and live counts Year and qualification mapping

Year 9 / Year 10 Lesson

Architecture Map

Retitle as a pathway architecture map and remove learner-facing year labels Y9/Y10 crosswalk and coverage data Curriculum Vault Map Add pathway, internal stage, qualification overlay and current status as separate fields Existing artifact identities Reference prototype lessons Remove year labels from visible title blocks and headers Internal filenames and curriculum tags Lesson-package templates Add variable 4–10 question rule and age-neutral

page furniture Version and stage codes Learner / Adult guidance templates Use v1.1 structures and standard worked-solution wording Lesson, unit and internal-stage metadata

15. Immediate Build Roadmap

1. Apply v1.1 to the remaining Foundations Pythagoras lessons and decide each next lesson count

from the mathematics and evidence.

2. Create a live Foundations coverage map showing Stepping Stones, Reconnection Routes, lesson

status and evidence.

1. Audit existing learner-facing lesson and guidance headers for year labels.
2. Update the earlier curriculum workbooks and maps in one coordinated versioned pass.

5. Build the Intermediate curriculum map against the selected IGCSE specification while retaining

the canonical NEO mathematical spine.

3. Confirm the Advanced qualification route and guided-learning model before fixing its timetable.

4. Continue testing lesson packages with NEO leadership and learner-facing review evidence.

16. Remaining Strategic Decisions

- Final IGCSE examination board and linear/modular route.
- Final AS/A-Level examination route and required Advanced guided hours.
- Where pathway names appear in the learner dashboard, navigation and progress reporting.
- Permanent Curriculum Vault storage for Scratchpad and other learner evidence.
- Final commissioner-facing reporting language for pathway progress without age-deficit framing.

17. Version History Update

v1.1 locks the Age-Neutral Pathway and Lesson Ecosystem Production Standard. It carries forward v1.0 Practice Companion Worked-Solution Unlock, v0.9 Scratchpad Design Rules, v0.8 Visual Geometry Containment and Rendering QA, v0.7 Nutrition as Explicit Mathematical Nourishment, v0.6 Learner-Flow and Interactive Entry Design, v0.5 Detailed Unit Blueprint decisions, v0.4 Lesson Support and Scratchpad Standard, v0.3 Guidance Layers, v0.2 Lesson Style and Language, and v0.1

Core Direction.

18. Planning Evidence Sources

Pearson Edexcel, A guide to modular International GCSEs: International GCSEs are described as two-year programmes with 120 guided learning hours.

<https://qualifications.pearson.com/content/dam/pdf/International%20GCSE/General/pearson-edexcel-guide-to-modular-international-gcse.pdf>

Cambridge International, Cambridge IGCSE Mathematics 0580 syllabus for 2025–2027:

approximately 130 guided learning hours, provided as guidance.

<https://www.cambridgeinternational.org/Images/662466-2025-2027-syllabus.pdf>

Cambridge International, AS & A Level Mathematics 9709 syllabus: approximately 180 guided learning hours for AS Level and 360 for the complete A Level.

<https://www.cambridgeinternational.org/Images/744603-2028-2030-syllabus.pdf>

Document control note — v1.1 milestone

This v1.1 milestone is retained as the complete previous locked record. The v1.2 additions and current document-control note follow on the next pages.

Internal classification: NEO Mathematics curriculum governance and production standard.

19. Locked One-Lesson-at-a-Time Design and Production Workflow

Version 1.2 locks the workflow validated through the Foundations lesson on Rounding to Significant Figures. The purpose is to protect curriculum quality, learner predictability and neurodivergent access while allowing each mathematical idea to receive an original, topic-appropriate interactive.

Status: LOCKED for all new NEO Mathematics lesson builds unless deliberately revised in a later GreenPrint version.

19.1 Stage 1 — Frame one lesson only

- Confirm the pathway, unit, lesson number, Stepping Stone, prerequisite knowledge and likely Reconnection Routes.
- Identify the mathematical noticing goal: the relationship, structure, change or decision the learner should be able to perceive.
- Work on one lesson at a time. Do not pre-build the remaining lessons in a topic before the current lesson has passed review.

19.2 Stage 2 — Present interactive options before building

Before any lesson is built, provide a short option bank of original or meaningfully adapted interactives. Normally present three to five viable options. For each option, state:

- the learner action and the mathematical response generated by the interactive;
- the Cornerstone it most strongly serves and how it can connect with the other five Cornerstones;
- the likely sensory and cognitive load, including any regulation or access considerations;
- the most useful Scratchpad background or representation;
- the likely Practice Companion focus and evidence opportunity; and
- a clear recommendation with a mathematical and accessibility rationale.

The lesson is not built until the curriculum lead chooses an option or authorises the recommendation. The chosen interactive may serve one Cornerstone most strongly, but it does not replace the requirement for all six Cornerstones to be present in the completed lesson.

19.3 Stage 3 — Use the reference shell as the design contract

The accepted Pythagoras lesson shell is the recurring learner-facing design contract. Sudden changes in layout, navigation, colour, control position or support sequence can create unnecessary demand for neurodivergent learners. Novelty should therefore come from the mathematics and the interaction, not from an unpredictable interface.

- Keep the familiar age-neutral title area, Driving Question, warm information block and compact support area near the top.
- Keep six separate collapsible Cornerstone panels in this recurring order: Connection, Movement, Reflection, Creativity, Rest and Nutrition.
- Keep the Final Task, Practice Companion and optional Scratchpad in their established positions and use the established organic palette, typography, spacing and control style.

- Give every Cornerstone an honest mathematical role in every lesson. The six panels need not receive equal time or weight.
- Use Nutrition through authentic applied context, intellectual nourishment or a brief sustaining-learning connection; do not force a contrived food context.
- Place explicit, step-by-step learner instructions, a noticing goal and a conceptual bridge inside each relevant panel.
- Do not add learner-facing chronological year labels. Internal stage tags may remain in filenames, metadata and curriculum maps.

Any proposed change to the shared shell must be reviewed as a system-level design decision, versioned in the GreenPrint and then applied consistently. A single lesson must not introduce an unannounced new shell.

19.4 Stage 4 — Build the complete lesson ecosystem

Once the interactive choice is approved, build the connected package together:

- Interactive Lesson with all six Cornerstones and a portfolio-capable Final Task;
- Practice Companion with 4–10 questions selected according to cognitive weight, representation variety and genuine practice need;
- optional lesson-local NEO Maths Scratchpad with a topic-appropriate background and the locked toolbar order;
- Learner Guidance in the locked age-neutral, calm and invitational structure; and
- Supporting Adult Guidance that protects learner thinking, regulation and agency without assuming specialist mathematical knowledge.

Timings remain invitations rather than deadlines. The learner may pause, return, continue in another session or use paper, a physical whiteboard, a notes app or another representation.

19.5 Stage 5 — Apply the worked-solution safeguard consistently

For every Practice Companion question independently, the Worked solution control begins visible but greyed out and disabled. It unlocks only after the learner has used the Socratic prompt, used the Hint and entered and checked two non-empty answers. The state is stored separately for each question and is not inherited from another question.

- A locked control states exactly which condition remains.
- Correct checking may confirm success without bypassing the support-pathway state for another question.
- A Supporting Adult does not provide the complete worked solution verbally while it remains locked.
- Navigation testing confirms that every question retains its own prompt, hint and attempt state.

19.6 Stage 6 — QA against the reference and review before progression

A lesson is not ready for review until the production check confirms:

- the visual shell and recurring control positions match the reference lesson;
- all six Cornerstones are visible, mathematically honest and presented in the locked order;
- learner-facing titles, headers, guidance and page furniture are age-neutral;
- every interactive responds meaningfully to learner input and accepts valid equivalent answers or tolerances;

- the Practice Companion question count is appropriate and the worked-solution rule passes independently for every question;
- the Scratchpad background persists while learner marks can be erased, undone, redone, saved locally and exported;
- guidance links, Practice Companion navigation, responsive layout and interaction extremes work after the latest change; and
- the source package, live preview, coverage map and evidence register carry the same version and status.

The curriculum lead reviews the live lesson. Requested changes are applied before the next lesson begins. Once accepted, the lesson becomes the evidence base for the next option bank; it does not silently change the shared shell.

20. Current Application — Accuracy, Powers and Scale

Lesson 1 — Rounding to Significant Figures: reference-shell version v0.2 accepted as the validation example for this workflow. It contains all six Cornerstones, an original Precision Lens and Boundary Explorer, a six-question Practice Companion, a place-value Scratchpad and two age-neutral guidance documents.

Lesson 2 — Estimation: begin with an interactive option bank and recommendation. Build only after the curriculum lead selects an option or authorises the recommendation. Use the same reference shell, ecosystem, support sequence, QA gate and evidence update.

21. Version History Update

v1.2 locks the Repeatable Lesson Workflow and Reference-Shell Consistency Standard. It carries forward v1.1 Age-Neutral Pathways and Lesson Ecosystem Production Standard, v1.0 Practice Companion Worked-Solution Unlock, v0.9 Scratchpad Design Rules, v0.8 Visual Geometry Containment and Rendering QA, v0.7 Nutrition as Explicit Mathematical Nourishment, v0.6 Learner-Flow and Interactive Entry Design, v0.5 Detailed Unit Blueprint decisions, v0.4 Lesson Support and Scratchpad Standard, v0.3 Guidance Layers, v0.2 Lesson Style and Language, and v0.1 Core Direction.

Document control note — v1.2

This v1.2 GreenPrint is the current locked running design record. Later decisions should be added through a new version rather than silently replacing this record.

Internal classification: NEO Mathematics curriculum governance and production standard.

22. Locked Reconnection Routes Access and Return Standard

Version 1.3 restores and locks the Reconnection Routes control as recurring page furniture in every NEO Mathematics lesson. The control sits near the top of the lesson beside Learner Guidance, Supporting Adult Guidance, the Maths Scratchpad and the Practice Companion.

Status: LOCKED for all new lesson builds and required in the audit of recent lesson packages.

22.1 Purpose

A Reconnection Route provides a dignified, targeted way to revisit a prerequisite that is materially blocking entry to the current mathematical task. It is attached to the moment of need and is not a separate catch-up course, punishment or deficit label.

22.2 Required learner-facing behaviour

- The Reconnection Routes button is visible in the compact support area near the top of every lesson.
- Selecting it opens a small in-page choice of only the prerequisites relevant to the current lesson.
- Each route names one short mathematical connection, offers a calm prompt and optional hint, and asks only for the minimum evidence needed to resume.
- Every route includes an explicit Return to [current lesson] control and preserves the learner's place.
- The learner may choose a route independently; a Supporting Adult may invite one without requiring it.
- Language remains age-neutral, non-deficit and invitational. Using a Reconnection Route is presented as part of learning, not evidence of failure.

22.3 Adult and evidence rule

The adult identifies the specific blocking prerequisite, invites the shortest relevant route, allows processing time and records only the minimum return evidence. The learner then returns to the original task. Reconnection activity must not replace the lesson's main evidence or become an additional completion burden.

22.4 Production and QA gate

- Confirm the button is present, keyboard reachable and located consistently with the other support controls.
- Confirm that every listed route is genuinely prerequisite to the current lesson and that no generic catch-up menu has been added.
- Test every prompt, hint, check and Return control.
- Confirm the route preserves the current lesson state and does not open an unrelated external course.
- Audit recent lesson packages and restore the control where it has been dropped.

22.5 Current validation example

Accuracy, Powers and Scale Lesson 2 — Estimation restores the control with four short routes: place value, rounding to significant figures, magnitude and the four operations. Each route remains inside the lesson and returns explicitly to Estimation.

23. Version History Update

v1.3 locks Reconnection Routes Access and Return. It carries forward v1.2 Repeatable Lesson Workflow and Reference-Shell Consistency, v1.1 Age-Neutral Pathways and Lesson Ecosystem Production, and all earlier locked GreenPrint decisions.

Document control note — v1.3

This v1.3 GreenPrint is the current locked running design record. Later decisions should be added through a new version rather than silently replacing this record.